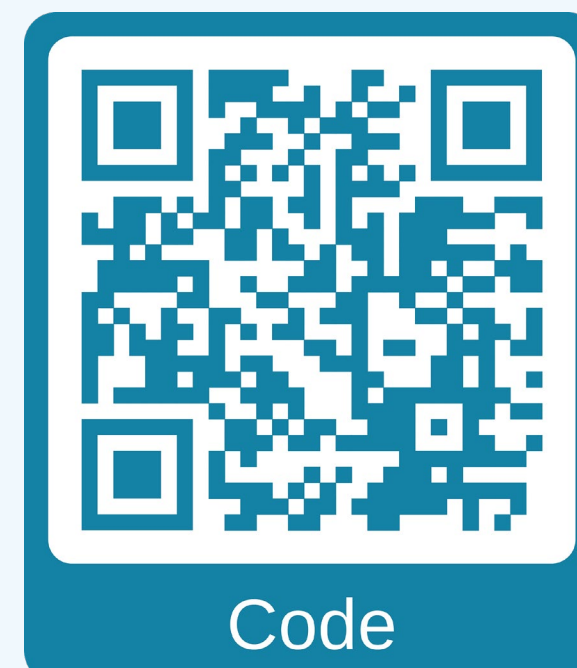
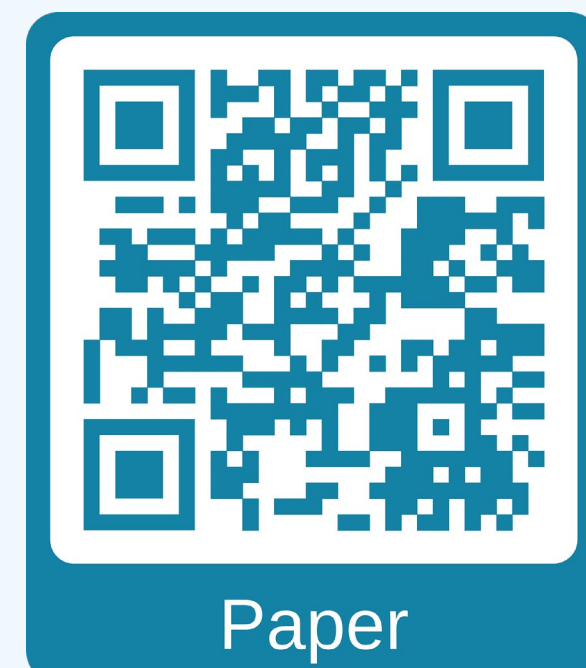
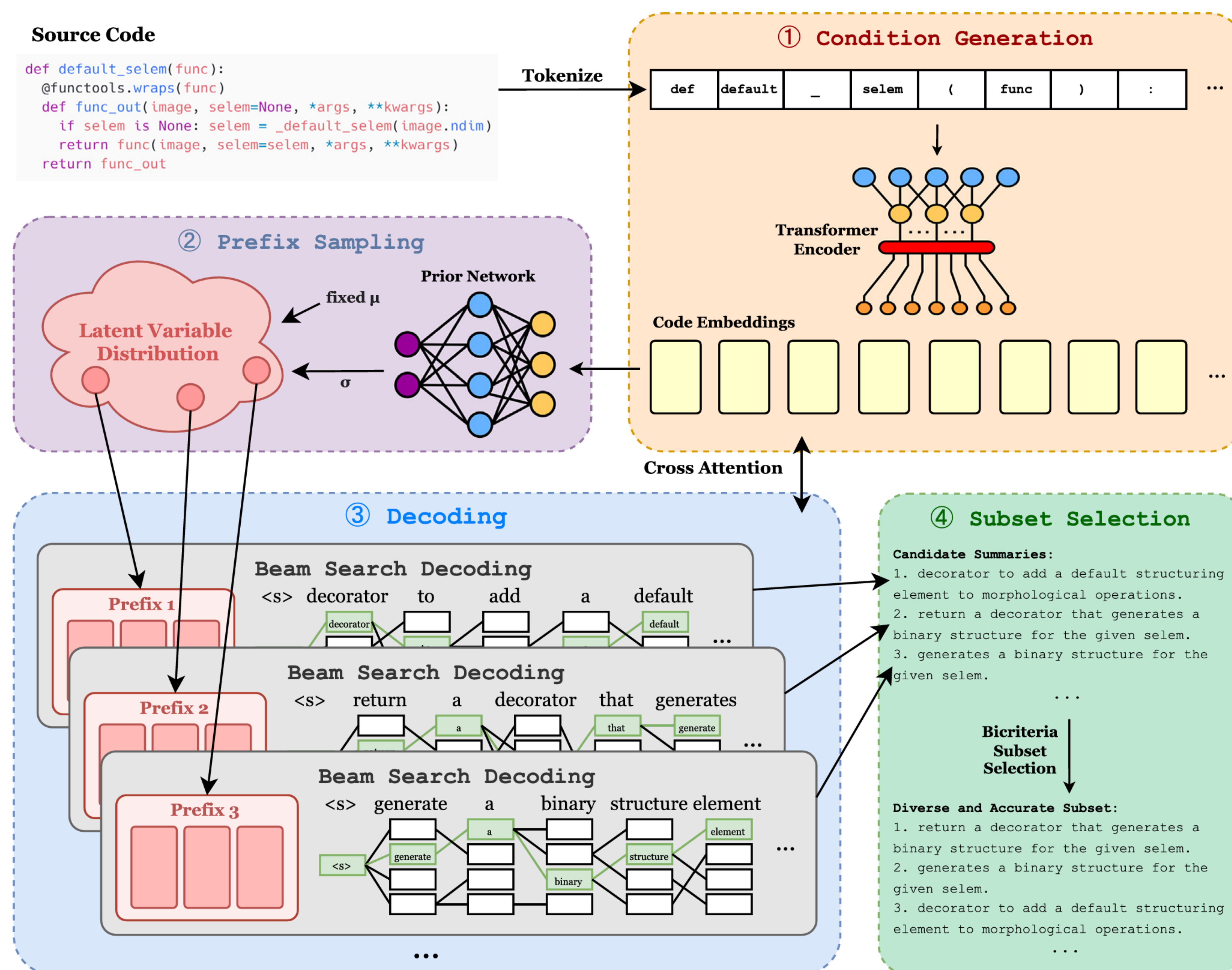


Highlights

- First work to enable diverse and accurate code summarization for Large Language Models of Code.
- Proposed a novel approach (VPT) to integrate the CVAE framework into LLM in a modular way, eliminating the need to retrain the entire LLM.
- Demonstrated the adaptability of VPT by applying it to several transformer-based pre-trained models.

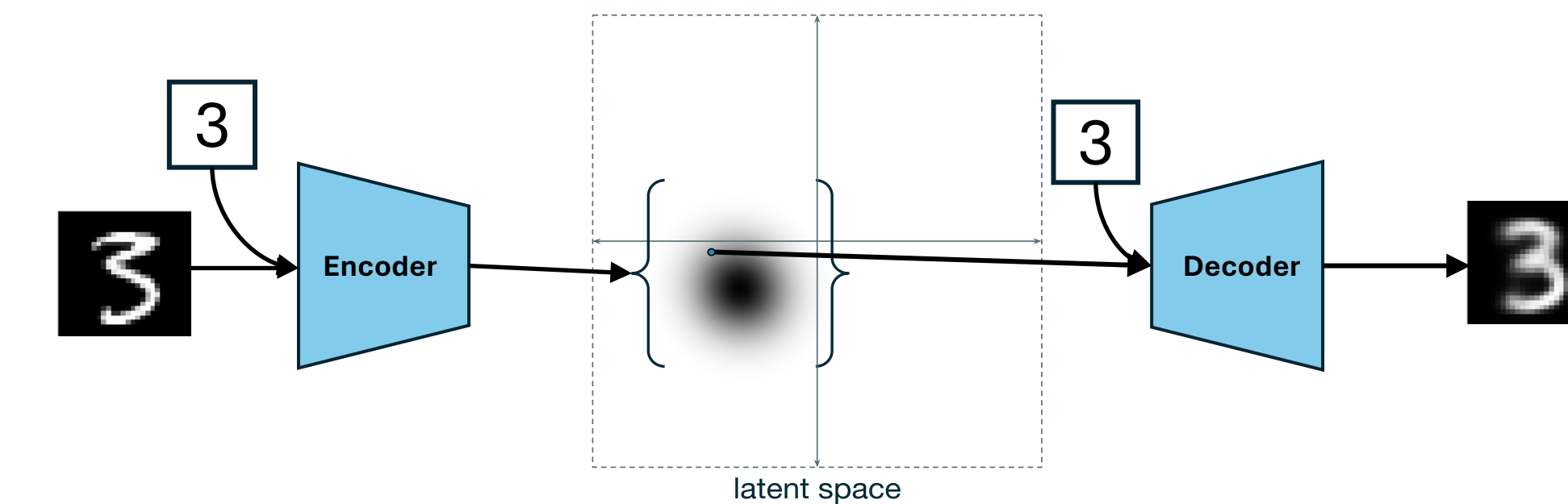


Generation Pipeline

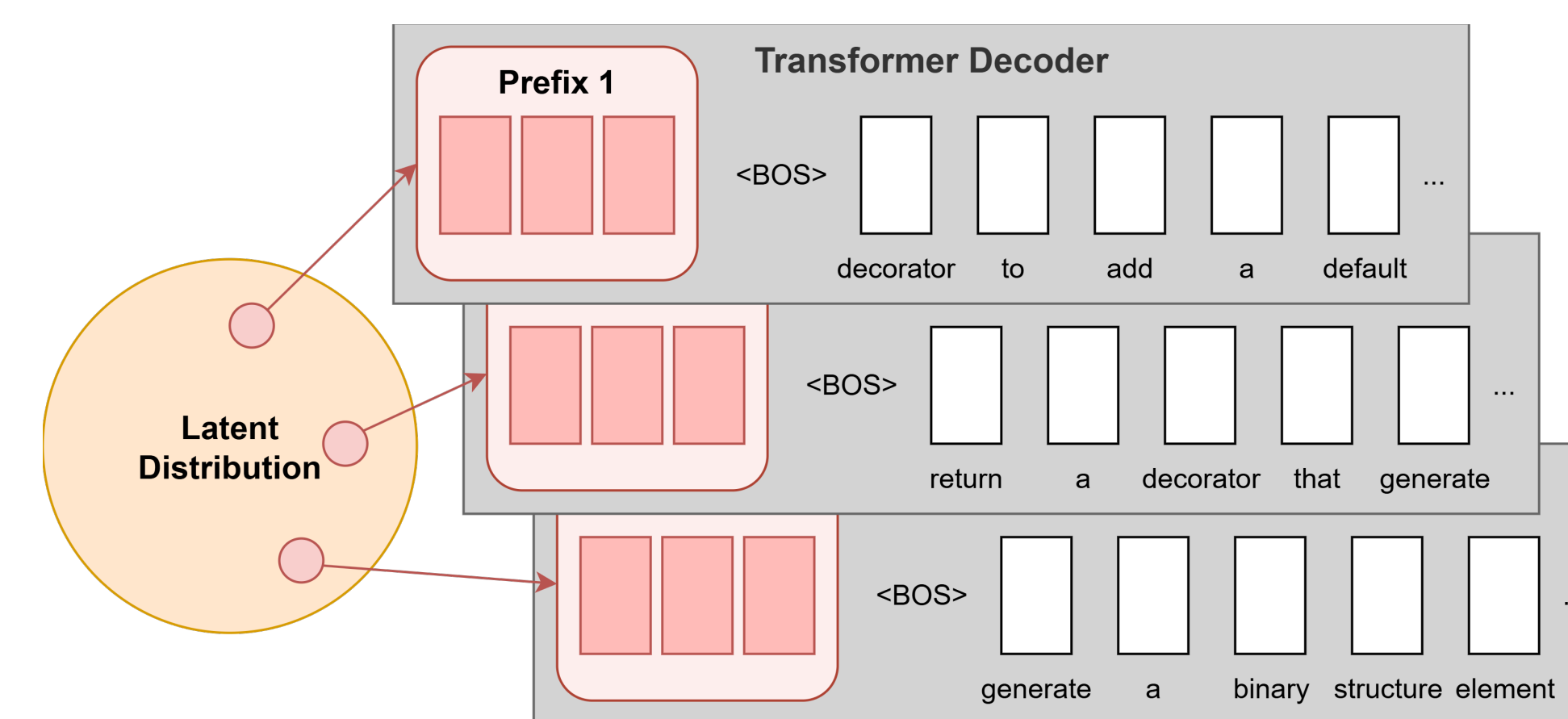


Motivation

- Conditional Variational AutoEncoders (CVAE) has been shown effective in other diverse generation tasks like dialog generation and image caption.



- Inspired by prefix tuning, We use a variable set of vectors sampled from a learned distribution to steer the decoding process into diversified outputs.



Diversity of Generated Summary

Method	#U	Python		
		D-1↑	D-2↑	S-B↓
Beam		16.79	33.02	79.63
Sampling		38.80	66.49	48.52
SBS	10	21.33	36.45	77.41
DBS		26.84	43.10	76.91
VPT		<u>28.36</u>	<u>51.97</u>	<u>59.05</u>
Beam		10.64	23.45	84.18
Sampling		31.90	58.69	54.98
SBS	20	13.61	26.25	83.15
DBS		11.95	21.54	86.20
VPT		<u>20.82</u>	<u>42.91</u>	<u>64.48</u>

Comparison of diversity in results generated by different methods

Code Summarization Performance

- We aim to maximize the likelihood of containing at least one adequate output. Therefore, we use the oracle metrics which takes the best result in the set.

Method	#U	B	R	M	BT-PR	BT-RE	BT-F1	SIDE
Beam	10	44.28	61.50	<u>42.14</u>	93.63	93.00	93.14	93.28
Sampling		<u>44.31</u>	<u>62.35</u>	41.50	<u>93.88</u>	<u>93.19</u>	<u>93.26</u>	<u>94.27</u>
SBS		42.30	59.76	40.50	93.25	92.76	92.80	93.33
DBS		43.07	59.90	40.53	93.28	92.58	92.81	92.54
VPT		46.40	63.83	43.52	94.16	93.38	93.62	95.21
Beam	20	45.63	63.12	<u>43.57</u>	93.85	93.33	93.40	94.53
Sampling		<u>45.84</u>	<u>64.08</u>	42.95	<u>94.27</u>	<u>93.50</u>	<u>93.58</u>	<u>95.16</u>
SBS		43.10	60.84	41.56	93.40	93.01	93.00	94.43
DBS		44.49	61.81	42.76	93.64	93.14	93.17	94.56
VPT		48.62	66.30	45.34	94.75	93.76	93.96	96.27
Beam	100	47.13	64.70	45.37	94.01	93.71	93.59	96.10
Sampling		48.92	67.25	45.77	94.97	93.95	94.15	95.97
VPT		51.86	69.83	47.75	95.78	94.23	94.60	97.18

Performance compared other diverse generation baselines in oracle metrics

Qualitative Examples:

Source code:

```
def default_selem(func):
    @functools.wraps(func)
    def func_out(image, selem=None, *args, **kwargs):
        if selem is None:
            selem = _default_selem(image.ndim)
        return func(image, selem=selem, *args, **kwargs)
    return func_out
```

Variational Prefix Tuning:

- decorator to add a default structuring element to morphological operations.
- generates a binary structure for the given selem.
- decorator to generate a binary structure of the image if no selem is specified.
- return a decorator that generates a binary structure for the given selem.
- return a binary structure for the given selem.
- decorator to generate a default structuring element for morphological operations.

Beam Search:

- decorator to add a default structuring element to morphological operations.
- decorator to add a default structuring element to morphology functions.
- decorator to generate a binary structure for binary morphological operations.
- decorator to generate a binary structure for an image if no selem is specified.
- decorator to generate a binary structuring element if no selem is specified.
- decorator to generate a binary structure for morphological operations.